DO NOW — RECALL

Boolean identities and De Morgan's laws

Name: _____ Class: _____ Date: _____

1. Using h(k)=k MOD 5, what is the starting bucket for 17?

2. What does an adjacency list store for a vertex?

3. What is the result of TRUE AND FALSE?

4. Simplify A ∧ 1.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

2.

2. Model answer

The neighbouring vertices.

3. Model answer

FALSE.

4. Model answer

A. AND with true leaves A unchanged.

Vocabulary — know these terms

Boolean identity — Rule for simplifying logical expressions.

De Morgan's law — Negate each operand and swap AND and OR.

complement — Logical NOT of a value or expression.

double negation — ¬¬A simplifies to A.

equivalence — Two expressions give the same output for every combination of inputs.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check and extend

Double negation gives ¬A ∨ B. The two expressions have matching outputs.

IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Simplify ¬(A ∧ ¬B) without drawing a gate diagram.

De Morgan gives ¬A ∨ ¬(¬B).

SIMPLIFIED

A B	Result
00	**1**
01	**1**
10	**0**
11	**1**

SIMPLIFIED truth table

NEW LEARNING

Read, annotate and explain

Negate the inputs

Distribute NOT across the brackets by negating A and B separately.

Swap the operator

When applying De Morgan's law, AND changes to OR; OR changes to AND.

Check equivalence

Both expressions produce the same output for all four combinations of A and B.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

NOT(A AND B)

A B	Result
00	1
01	1
10	1
11	0

NOT(A AND B) truth table

New Learning

NOT A OR NOT B

A B	Result
00	1
01	1
10	1
11	0

NOT A OR NOT B truth table

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

De Morgan's law allows us to remove a NOT from brackets without changing AND or OR.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Simplify ¬(A ∨ ¬B) and justify the final result.

I DO — WORKED EXAMPLE

Follow the worked method

Simplify ¬(A ∨ B) and verify its truth table.

Predict and explain

Negate A and B separately and replace OR with AND.

Check and extend

Both expressions are true only when A=0 and B=0, proving equivalence.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Simplify ¬(A ∨ B) and verify its truth table.

Negate A and B separately and replace OR with AND.

NOT(A OR B)

A B	Result
00	**1**
01	**0**
10	**0**
11	**0**

NOT(A OR B) truth table

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Simplify A ∨ 0.

Question 2

Use De Morgan to simplify ¬(P ∧ Q).

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict and explain De Morgan gives ¬A ∨ ¬(¬B).

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Simplify ¬(A ∨ B) and verify its truth table.

Negate A and B separately and replace OR with AND.

NOT A AND NOT B

A B	Result
00	1
01	0
10	0
11	0

NOT A AND NOT B truth table

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Simplify ¬(A ∧ ¬B) without drawing a gate diagram.

De Morgan gives ¬A ∨ ¬(¬B).

ORIGINAL

A B	Result
00	**1**
01	**1**
10	**0**
11	**1**

ORIGINAL truth table